Stress Test of Candidate Analyses for the Biomarker-by-Drug Interaction in the Hybrid Design: Working Protocol

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Purpose. This working protocol sets out how the project chooses its analysis for the biomarker-by-drug interaction in the Hybrid design. The study is exploratory: the protocol was revised as results came in (Amendments 1 and 2), and its findings should be confirmed in an independent study.

It names a *strawman*: a provisional analysis, put forward to be attacked. It defines four criteria the analysis must meet: nominal, robust, powerful and sensitive. It lists the challengers, and a set of stress conditions designed to break the strawman. It then states the rule by which the strawman survives or is replaced.

The strawman is chosen on the evidence in `docs/45`, read under the strict size criterion of Section 3. It is not the primary analysis recommended there. Section 8 explains the change.

The study follows the ADEMP structure: aims, data-generating mechanisms, estimands, methods, performance measures (Morris, White and Crowther [1]).

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md` and `docs/45`. Symbols marked *local* are defined here.

Table 1: Notation index and glossary of symbols used in this document

Symbol	Meaning	Status
Y_{it}	symptom score (code `Sx`); $t = 0$ is baseline	canonical
BR_{it}, PB_{it}, TV_{it}	drug, expectancy and natural-course responses	canonical
B_i, b_i	biomarker; standardized biomarker	canonical
D_{it}, $D_{bc,it}$	binary drug state; exposure-decayed indicator, 0 before any exposure	canonical
$t_{1/2}$, $t^a_{1/2}$	true carryover half-life; half-life assumed by the analysis	canonical; local (`docs/45`)
c_{bm}	biomarker-BR coupling (the predictive effect)	canonical
$c_{bm,TV}$, $c_{bm,PB}$	biomarker-TV and biomarker-PB correlations (prognostic couplings)	local; $c_{bm,PB}$ is the package's `c.bm.pb`
β^{TV}_{bm}	mean moderation of the natural course by b_i	canonical (component slope)
β_B, $\beta_{bm:D}$	biomarker main effect; the interaction, the estimand	local (`docs/41`); canonical
ρ	AR(1) serial correlation per week	canonical
$\sigma_{\varepsilon,\mathrm{OL}}$, $\sigma_{\varepsilon,\mathrm{bl}}$	SD of added measurement error, open-label and blinded visits	local
κ	mean model SE over empirical SD	canonical
π	power	canonical
N	total sample size	canonical
α	nominal level, 0.05	canonical; row added 2026-10-07
η_t	expectancy at visit t (code `De`): 0 baseline, 1 open label, 0.5 blinded	canonical (NOTATION.md)
σ_{TV}, $G_{TV}(t)$	SD and Gompertz mean curve of the natural-course component	local; row added 2026-10-07

Symbol	Meaning	Status
k	Weibull shape of the decay family	canonical; row added 2026-10-07
$c_{bm,BL}$	biomarker-baseline correlation (baseline-severity prognostic coupling; Amendment 2)	local; row added 2026-10-07

Glossary.

- **Strawman.** A provisional analysis put forward to be criticized and tested, not a deliberately weak one. In `docs/36` and `docs/44` the word named the published-code reference arm; that arm is called the *published reference arm* here.
- **Challenger.** An analysis that could replace the strawman.
- **Stress cell.** A condition chosen because it is expected to break at least one candidate.
- **Power curve.** Size-adjusted power as a function of the true coupling c_{bm}, the basis of the sensitivity criterion.
- **Base cell.** The condition from which each stress varies one factor.
- **Prognostic biomarker.** A biomarker associated with the outcome regardless of treatment, here through TV or PB.

1. Aims

1. Test whether the strawman (Section 4) holds its nominal size in every stress cell.
2. Compare its power and sensitivity with every challenger that also holds its size.
3. Replace the strawman if, and only if, the rule of Section 7 says so.
4. Report the published reference arm alongside, for comparability with Hendrickson et al. [2].

2. Scope

- **Design:** Hybrid only. Visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20, plus baseline. Four paths with the published 18/18/17/17 allocation at $N = 70$, scaled proportionally at other N.
- **Estimand:** $\beta_{bm:D}$.
- **Test:** two-sided at $\alpha = 0.05$.

3. Criteria

Nominal. Two conditions must both hold:

- the pooled rejection rate over all null cells lies within 2 Monte Carlo SE of 0.05;
- no null cell exceeds 0.075.

Each null cell has 1,000 replicates, so the MCSE is 0.0069 per cell. There is no "slightly liberal" class. An analysis that fails either condition fails the criterion.

Robust. Nominal in every stress cell of Section 5, each assessed separately with the cell condition above. A failure in a single stress cell is recorded with the condition that caused it.

Powerful. Power is size-adjusted: each analysis rejects at the 5% quantile of its own p-values in the matching null cell. The summary is the maximin shortfall. In each alternative cell, compute each analysis's shortfall from the best nominal analysis in that cell. The summary for an analysis is its largest shortfall over the cells. Unadjusted power is also reported.

Sensitive. The test should respond to the strength of the true interaction: power should rise steadily as the true coupling c_{bm} moves away from zero in either direction, and reach α at zero. It is assessed on the power curve, size-adjusted, over the grid of Section 5. Its components are:

- **monotonicity**: no decrease in power between adjacent grid points larger than 2 MCSE, on either side of zero;
- **symmetry**: power at $-c_{bm}$ within 2 MCSE of power at $+c_{bm}$;
- **steepness**: the average slope of power against $|c_{bm}|$ between 0.15 and 0.30;
- **detectable coupling**: the smallest $|c_{bm}|$ at which power reaches 0.80, by linear interpolation on the grid. Smaller is more sensitive.

The curve is estimated in two constructs. In the base construct (AR(1)) $|c_{bm}|$ runs to 0.45. In the compound-symmetry construct it runs to 0.25, the largest value below that construct's ceiling of 0.256.

Estimand fidelity is also reported, though it is not a selection criterion: the relative bias of $\widehat{\beta}_{bm:D}$ against the value the same analysis gives at $t_{1/2} = 0$, when $t^a_{1/2}$ is close to $t_{1/2}$.

4. The strawman and the challengers

Strawman S. Random intercept plus `corCAR1` residuals, with CR2 standard errors and Satterthwaite degrees of freedom (Bell and McCaffrey [3]). Baseline

is a response row, coded off drug. The model uses linear time and $D_{bc,it}$ at $t^a_{1/2} = 0.5$ week:

```
nlme::lme(Sx ~ bmc + t + Dbc + bmc:Dbc, random = ~ 1 | ptID,
          correlation = nlme::corCAR1(form = ~ t | ptID))
# then clubSandwich::coef_test(fit, vcov = 'CR2',
#                              test = 'Satterthwaite')
```

It is chosen because it is the most powerful analysis that was nominal, under the strict criterion, in all eight null cells of `docs/45`. Its sizes there were 0.040 to 0.064, with a mean of 0.052.

Challengers. All are fitted to the same trials.

Table 2: Challenger analyses with identifiers and reasons for inclusion

ID	Analysis	Reason
A	unstructured (`us`), Kenward-Roger, baseline row, linear t, $D_{bc,it}$ at 0.5	primary of `docs/45`; most powerful under AR(1) there
C1	S with `nlme::varIdent()` residual variances by `phase`, phase = baseline / open label / blinded; CR2	gives the parsimonious model the phase heteroscedasticity that likely drives A's advantage
C2	C1 with model-based inference (the `nlme` Wald t-test; `nlme` has no Satterthwaite option for this model)	whether CR2 costs power once the variance model is closer
DS	S with $t^a_{1/2}$ chosen by AIC over {0, 0.25, 0.5, 1, 2} (0 meaning D_{it}); ML for selection, REML refit	data-chosen half-life
DA	A with the same AIC rule	the same, for the unstructured model
ES, EA	S and A with β_B freed at baseline (`+ bmc:bl0`); both are fitted, so the comparison is not chosen after the results	robustness to failure of the constancy assumption

ID	Analysis	Reason
F1	S plus `De + bmc:De`, where `De` is the expectancy (1 open label, 0.5 blinded, 0 at baseline)	remedy for a placebo-prognostic biomarker
F2	S plus `bmc:t`	remedy for a natural-course-prognostic biomarker
R	published reference arm: random intercept, model-based, linear t, binary D_{it} (`cs-Satt` in `docs/45`)	comparability

ES, EA, F1 and F2 change the model, not only the inference. They are evaluated in every cell, so their cost when nothing is wrong is measured as well as their benefit when something is.

5. Data-generating mechanisms

Base construct. Configuration C of `docs/36`:

- separable AR(1) within each component at $\rho = 0.7$ per week;
- graded coupling of the biomarker to BR;
- published Gompertz means and SDs;
- the biomarker coupled only to BR.

This is the construct `docs/44` identifies as the best implemented. The base cell is:

- $N = 70$, $t_{1/2} = 0.5$ week, no dropout, exponential decay.

Power-curve cells (the sensitivity criterion):

Table 3: Power-curve cells by construct with their c_{bm} grids

Construct	Grid of c_{bm}
base (AR(1), $\rho = 0.7$)	−0.30, 0, 0.075, 0.15, 0.225, 0.30, 0.375, 0.45
compound symmetry, 0.8 at every lag	−0.20, 0, 0.05, 0.10, 0.15, 0.20, 0.25

The zero cells have 1,000 replicates, all others 500. A negative c_{bm} means the biomarker is coupled to a smaller drug response. Under graded coupling the joint distribution exists for negative values on the same terms as for positive ones, which is checked before the run. The base-construct cells at 0 and 0.30 double as the null and alternative of the base cell.

Stress cells. Each varies one factor from the base cell. Each has a null twin ($c_{bm} = 0$, 1,000 replicates) and an alternative at $c_{bm} = 0.30$ (500 replicates), unless the table says otherwise.

Table 4: Stress-test factors, levels and the analyses each is expected to break

#	Factor	Level	Expected to break
1	correlation	compound symmetry, 0.8 at every lag (`docs/45` CS): covered by the compound-symmetry power curve, whose largest value is 0.25 because 0.30 exceeds that construct's ceiling	pure-AR(1) assumptions; tests S under its weakest structure
2	correlation	base plus measurement error, $\sigma_{\varepsilon,\mathrm{OL}} = 3$, $\sigma_{\varepsilon,\mathrm{bl}} = 6$	homoscedastic working models (S, F1, F2)
3	sample size	$N = 35$	`us` with Kenward-Roger (A, DA); CR2 degrees of freedom
4	sample size	$N = 150$	none expected; shows whether any excess persists with N
5	carryover	$t_{1/2} = 0$	$D_{bc,it}$ when there is no carryover
6	carryover	$t_{1/2} = 1$ week	binary D_{it} (R); short $t^a_{1/2}$
7	decay form	Weibull, $k = 0.5$ (heavy tail)	the exponential $D_{bc,it}$

#	Factor	Level	Expected to break
8	decay form	Weibull, $k = 2$ (light tail)	the exponential $D_{bc,it}$
9	dropout	20% monotone, completely at random	covariance estimation (A)
10	dropout	20% monotone, at random given the last observed Y	all; MAR is assumed by every likelihood analysis
11	prognostic	$c_{bm,TV} = 0.3$ (time-constant natural-course association)	the shared β_B of the row coding (S, A, C)
12	prognostic	$\beta_{bm}^{TV} = 0.3$ (association growing along the TV curve)	every analysis without `bmc:t`
13	prognostic	$c_{bm,PB} = 0.3$ (expectancy-scaled placebo association)	every analysis without `bmc:De`

For the prognostic cells 11 to 13, the alternative twin keeps the prognostic coupling as well. The question is whether the remedies recover the interaction, not only whether the null holds.

Implementation of the new mechanisms in the `04-power-simulation.R` construct. The biomarker is index 1 of the joint vector.

- $c_{bm,TV}$, $c_{bm,PB}$. Set the biomarker's correlation with every post-baseline TV (or PB) element to the stated value. The PB SD is already scaled by the expectancy η_t, so the implied slope is expectancy-scaled. If the matrix is not positive definite the cell stops with an error; no repair is permitted.
- β_{bm}^{TV}. After the draw, add $\beta_{bm}^{TV}\, b_i\, \sigma_{TV}\, G_{TV}(t) / \max_t G_{TV}(t)$ to TV_{it}. This is mean moderation that grows along the natural-course curve.
- **Measurement error.** Add independent normal error to Y_{it}, with SD 3 at open-label visits and 6 at blinded visits; none at baseline.
- **Weibull decay.** Applied to both the BR carryover and the graded coupling, as in manuscript 02's decay family. The analysis keeps the exponential $D_{bc,it}$.
- **Dropout.** Monotone, starting at a post-baseline visit:
  - MCAR: drawn uniformly over visits 3 to 8 for 20% of participants;

  - MAR: the hazard at each visit increases with the previous observed $Y_{i,t-1}$ (logistic, calibrated to 20% overall).

6. Methods and performance measures

Replicates: 1,000 per null cell and 500 per alternative cell, with per-chunk seeds as in `08-covariance-study.R`. All analyses are fitted to the same trials.

Reported for each analysis and cell:

- rejection rate with its MCSE;
- size-adjusted power;
- mean estimate, empirical SD, mean model SE and κ;
- relative bias where defined;
- the distribution of the selected half-life (DS, DA);
- failed fits.

Failed fits count as non-rejections, and their rate is reported. An analysis with more than 2% failed fits in any cell is flagged.

Paired comparisons. Differences in power between two analyses in a cell are tested with McNemar's test on the paired rejections.

7. Decision rule

1. **Screen.** Discard every analysis that fails the nominal criterion in any null cell, base or stress, except the prognostic stress cells 11 to 13.
2. **Prognostic screen.** Cells 11 to 13 are judged separately. An analysis that fails one of them is not discarded. It is flagged, and recommended only together with the remedy (ES, EA, F1 or F2) that is nominal in that cell.
3. **Sensitivity screen.** Flag any survivor whose power curve fails the monotonicity or symmetry condition in either construct; a flagged analysis may not be selected.
4. **Select.** Among the remaining survivors, select the analysis with the smallest maximin shortfall over the alternative cells, power-curve cells included. Report the detectable coupling and steepness of every survivor, and break ties within 0.02 of maximin shortfall by the smaller detectable coupling in the base construct.
5. **Strawman rule.** S is retained unless a survivor beats it on the maximin shortfall by more than 0.05, with the McNemar difference significant at 0.05 in at least half the alternative cells. Ties go to the analysis with fewer covariance parameters.

6. **Replacement.** If S is replaced, the winner is reported as the recommended analysis. It is not stress-tested again in this study, because doing so would make the selection adaptive.

Changes to these rules made after results were seen are recorded in the amendments below as revisions of the working protocol, each with its reason.

8. Relation to earlier documents

- **`docs/45` recommended A as primary.** Under the strict criterion of Section 3, A's mean size of 0.058 fails the nominal criterion (the 2-MCSE band is 0.043 to 0.057). The "slightly liberal" class of `docs/45` was defined after the results were seen. This document removes it, and S becomes the strawman.
- **The parsimonious challengers are given the feature A exploited.** In `docs/45` they were not. C1 adds the phase-specific variance that was the likely source of A's power advantage.
- **The constancy assumption is now tested.** `docs/45` could not test it, because every construct satisfied it. Cells 11 to 13 do.

9. Compute and execution

Size. 15 power-curve cells (8 base, 7 compound symmetry) plus 13 stress cells $\times$ 2, about 40 cells once the curve cells shared with the stress set are counted once. About 12 analyses per replicate, with about 24,000 replicates in total. A, DA and EA use unstructured Kenward-Roger fits and dominate the cost.

Time. The pilot (20 replicates per cell, 780 in all, 16.7 minutes on 8 cores, no failed fits) puts the full run at about 9 to 10 hours. An earlier estimate of four to five days, scaled from the exposure-coding pass of `docs/45`, was too high. The run is local and detached, with each chunk of 25 replicates checkpointed.

Order of work.

1. Implement the mechanisms of Section 5 and the analyses of Section 4 in a new driver, `11-strawman-stress-test.R`. Add unit checks:
   - the positive-definiteness check on every new matrix;
   - the prognostic couplings reproduce their target correlations in a large draw;
   - the dropout rates match their targets;
   - the AIC rule reproduces the fixed $t^a_{1/2}$ analysis when it selects it.
2. A pilot of 20 replicates per cell, to check every analysis fits and to time the run.

3. The full run.
4. A results document (`docs/47`), reporting against this working protocol, its revisions included.

10. Limitations

- One design (Hybrid) and one construct family (Gompertz components with separable AR(1) or CS).
- The prognostic couplings and the measurement-error SDs are chosen values, not estimates from trial data.
- Stress factors are varied one at a time, so interactions between stresses (for example small N with dropout) are not tested.
- The strawman's own analysis half-life (0.5 week) and the AIC grid are fixed here. Other values are not explored.

Amendment 1 (2026-10-07, after the results were seen)

Two revisions of the working protocol's decision rule, made after the full run. The original rule remains reproducible with the defaults of `12-strawman-stress-decision.R`.

1. **The symmetry screen is dropped.** Without prognostic coupling, power is exactly symmetric in the sign of c_{bm} by construction (`docs/48`, Section 4). Its 9 failures in 22 tests were Monte Carlo error, confirmed by a fresh-seed rerun. The monotonicity screen is kept.
2. **The strawman selects its half-life by AIC.** It is analysis DS: S with $t^a_{1/2}$ chosen over {0, 0.25, 0.5, 1, 2}, ML for selection and REML refit. All other rules are unchanged.

Run with `--strawman DS --no-symmetry`. Outputs carry the suffix `-ds-nosym`.

Outcome.

- **Survivors** of the nominal and monotonicity screens: S, C1, C2, DS, ES, EA, F1 and F2.
- **Prognostic screen:** S, C1, C2, DS and F2 are flagged. ES, EA and F1 are not.
- **The strawman is replaced.**
  - C1 beats DS on maximin shortfall by 0.148 (0.034 against 0.182), and is significantly ahead by McNemar's test in 19 of 22 alternative cells.
  - C2 beats it by 0.142, ahead in 20 of 22.

  - S itself beats DS by only 0.038. AIC selection costs a little power against the fixed half-life of 0.5 week.

- **The maximin winner is C1, but it is flagged** (null rejection 0.382 under a constant natural-course association). By rule 2 it may be recommended only together with a remedy that is nominal in that cell, and no such C1 variant was in the original protocol.
- **F1 is the best of the unflagged analyses** (maximin shortfall 0.242, against 0.278 for ES and 0.318 for EA).

The combination of C1 with F1's remedy is being run as a post hoc arm (`11-strawman-stress-test.R --extra`). It can motivate a recommendation but cannot replace this outcome without an independent run.

Amendment 2 (2026-10-07, after the results were seen)

This amendment records four further revisions of the working protocol, all post hoc, and the outcome of the decision rule rerun with every analysis. The original protocol text above, and Amendment 1, are unchanged in substance. Two edits to them are typographical: the C1 row of the challenger table showed a stray backslash in its code span and was reworded to name `nlme::varIdent()` by `phase`; and the expectancy symbol e_t was renamed η_t, the canonical symbol of `analysis/report/NOTATION.md`, with no change of meaning.

1. Two baseline-severity cells (40 and 41). The biomarker is correlated $c_{bm,BL} = 0.3$ with the baseline level BL_i, and with no response component: a null twin (cell 40, 1,000 replicates) and an alternative at $c_{bm} = 0.30$ (cell 41, 500). Such an association is the same at every visit, so the constancy assumption holds. The cells are judged as prognostic cells under rule 2, alongside cells 11 to 13. The study now has 41 cells, 15 null and 26 alternative, and 28,000 simulated trials.

2. Three analyses (G1, G2, F1A).

- **G1:** the F1 mean model with phase-specific residual variances (`nlme::varIdent()` by phase, as in C1), CR2 inference;
- **G2:** the same model with model-based inference, as in C2;
- **F1A:** F1 with $t^a_{1/2}$ chosen by AIC over $\{0, 0.25, 0.5, 1, 2\}$, ML for selection and REML refit, as in DS.

They are fitted to the same trials and judged by the same rule as the original analyses. Being post hoc, none can replace the outcome of Amendment 1 without an independent run.

3. Failed fits. Section 6 states that failed fits count as non-rejections. The summaries of `11-strawman-stress-test.R` and the decision script instead exclude them from the denominator. Failed fits number 0 to 12 per analysis in

28,000 trials, at most 6 of 1,000 (0.6%) in any cell (G1 and G2), below the 2% flag. Counting them as non-rejections would lower a cell's rejection rate by at most 0.0025 (verified, `summary-full*.csv`). This is far smaller than the margin of any size screen outcome, so no outcome depends on the convention (inferred from the margins).

4. Outcome with all 14 analyses (verified; the added analyses come from `11-strawman-stress-test.R --extra` and `--extra2`, and `12-strawman-stress-decision.R --strawman DS --no-symmetry --with-extras` writes `decision-*-ds-nosym-all14.csv`).

- **Size screen.** A (0.057 pooled; 0.087 at $N = 35$), DA (0.065), R (0.069) and F1A fail. F1A rejects 0.057 pooled, outside the 2-MCSE band of 0.046 to 0.054, and 0.073 in the $t_{1/2} = 1$ cell. AIC selection therefore does not hold its size once the expectancy terms are in the model.
- **Survivors** of the size and sensitivity screens: S, C1, C2, DS, ES, EA, F1, F2, G1 and G2.
- **Baseline-severity cells.** In the null cell every analysis is within the cell limit, 0.040 (F1) to 0.075 (R), as the constancy argument predicts; S rejects 0.050 and A 0.051.
- **Prognostic screen.** The analyses nominal in all four prognostic null cells are ES, EA, F1, G1 and G2 (G1 0.042 to 0.055, G2 0.043 to 0.062). S, C1, C2, DS and F2 remain flagged.
- **Maximin shortfall** over the non-prognostic alternative cells, among the prognostic-robust survivors: G2 0.232, F1 0.242, G1 0.252, ES 0.278, EA 0.318. C1 has the smallest value of all survivors (0.034) but is flagged.
- **Selection.** G2 and F1 tie within 0.02. The tie-break of rule 4, the detectable coupling in the base construct, is undefined, because neither reaches power 0.80 on the grid. By the tie rule of rule 5 (fewer covariance parameters), F1 is selected. The outcome of Amendment 1 is unchanged.
- **Over all 26 alternative cells,** prognostic cells included, the order among the robust survivors is G1 0.308, G2 0.312, F1 0.358, EA 0.528, ES 0.544. On this reading the phase-variance versions lead F1 by about 0.05, mainly through the natural-course prognostic alternatives. Because the protocol does not say whether the prognostic alternative cells count (`docs/48`, Section 3), and G1 and G2 are post hoc, this is recorded and does not change the selection.

Size-adjusted power of the added analyses in the two reference cells: at $c_{bm} = 0.30$ in the AR(1) construct, F1 0.364, G1 0.354, G2 0.380 and F1A 0.338; at $c_{bm} = 0.25$ under compound symmetry, F1 0.522, G1 0.546, G2 0.528 and F1A 0.446 (verified, `decision-power-ds-nosym-all14.csv`).

11. References

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